

Cadence × Lightmatter — High-Speed Analog F2F

Agenda Outline

Proposed Date: Week of August 10, 2026 **Format:** In-person working session, ~full day

Attendees: Analog design leads and architects from both teams

Objective

Align both teams on how high-speed signals interact at the integration boundary between LM's analog front-end IP (TIA and Driver) and the Cadence SerDes macro. The link is fully analog — no ADC or DSP — so both teams must reach precise agreement on signal contracts, simulation methodology, and system-level interactions.

Agenda

#	Topic	Led By	Time
1	Introductions & Program Context	Both	15 min
2	Architecture Overview — Signal Path Walkthrough	LM	15 min
3	Rx TIA design overview	LM	60 min
4	Surrounding Rx circuitry (buffer/sampler/slicer)	CDNS	60 min
5	Lunch/Demo	Both	60 min
6	Tx Serializer (Tx core)	CDNS	60 min
7	Tx Driver design overview	LM	60 min

#	Topic	Led By	Time
8	Open Items, Action Assignments & Next Steps	Both	15 min

Topic Descriptions

1. Introductions & Program Context Brief team introductions and a shared statement of program goals, schedule pressure, and the scope of the co-development relationship. Confirm attendee roles and establish who the technical decision-makers are for each team.

2. Architecture Overview — Signal Path Walkthrough LM walks the full signal path from PD through TIA to SerDes Rx, and from SerDes Tx through the driver to MRM.

3. Rx Boundary — TIA design deep-dive (LM) LM presents the design(s) of the TIA / CTLE / amplifier chain that drives the Rx boundary: topology, bandwidth, gain, output swing and common-mode, input-referred noise, testability and the DC offset cancellation (DCOC) loop.

4. Rx Boundary — Surrounding Rx circuitry (CDNS) CDNS presents the buffer, S/H and slicer which the TIA output drives: topology, bandwidth, gain, output swing and common-mode, input-referred noise and testability

5. Tx Boundary — Surrounding Tx circuitry (CDNS) CDNS presents the Tx core feeding the driver(s): output format and slew rate, tap weight control ownership, and PVT interactions across the boundary. Goal: a mutually agreed interface specification at this boundary.

6. Tx Boundary — Analog driver design deep-dive (LM) LM presents the driver: pre-emphasis implemented entirely in the analog domain (no digital FIR), driver input requirements,

Cross-cutting themes (discussed throughout sessions 3–6, in the context of each circuit)

- *Co-simulation & model exchange:* Verilog-A behavioral models from both teams, a shared Virtuoso interface harness at each boundary, joint sign-off etc etc.

7. Open Items, Action Assignments & Next Steps Capture all unresolved items from the day, assign owners and due dates, and agree on the next joint checkpoint.